

2082

Bachelor of Science in Computer Science and Information Technology

Course Title: **Computer Graphics**

Code No: **CSC214** Semester: **3rd Semester**

Full Marks: **60** Pass Marks: **24** Time: **3 hours**

Candidates are required to answer the questions in their own words as far as practicable.

Section A

1. Explain Bresenham's line drawing algorithms. Compare it with DDA line algorithm. Draw a line from (2, 3) to (10, 8) by using Bresenham's algorithm.
2. What is a polygon mesh? Describe its types. Construct a polygon table and edge table for a cube of side 2 units placed with one vertex at origin.
3. Explain the Painter's algorithm for visible surface detection. Explain the BSP tree method used for visible surface determination. How does it divide the space and organize objects in a scene?

Section B

Attempt any Eight questions[8×5=40]

4. Explain the need for machine-independent graphics languages. How do such standards benefit application developers?
5. Define 2D rotation in computer graphics. Derive the rotation matrix and calculate the new coordinates of a point (2, 3) after a rotation of (45°) about the origin.
6. How is the transformation matrix computed when switching from one coordinate system to another? Illustrate with an example.
7. Explain the 3D viewing pipeline in computer graphics. Explain about how a 3D world coordinate system is transformed to a 2D screen?
8. For control points $P_0(0,0)$, $P_1(1,2)$, $P_2(3,3)$, and $P_3(4,0)$, calculate the Bezier curve point at $u = 0.5$. Also plot the rough curve shape.
9. What is spatial-partitioning representation? Explain how it differs from boundary representation in terms of geometry storage and processing.
10. What is constant intensity shading? Compare Phong shading and fast Phong shading.
11. Discuss the use of virtual reality in education. How does VR enhance student engagement and learning outcomes?

12. Write short notes on a. Lighting in OpenGL. b. Orthographic projection.